

EXPERIMENT 15

AIM

Monitor a directory for a specified period and record created, modified and deleted files with timestamps.

ALGORITHM

1. Specify the directory to monitor.
2. Take an initial file snapshot.
3. Wait for the observation interval.
4. Take a second snapshot.
5. Find created and deleted files by comparing filenames.
6. Find modified files by comparing timestamps.
7. Display each activity with a timestamp.
8. Stop.

CODE

```
import os
import time
from datetime import datetime

folder = "monitor"

def snapshot():
    result = {}
    for name in os.listdir(folder):
        path = os.path.join(folder, name)
        if os.path.isfile(path):
            result[name] = os.path.getmtime(path)
    return result

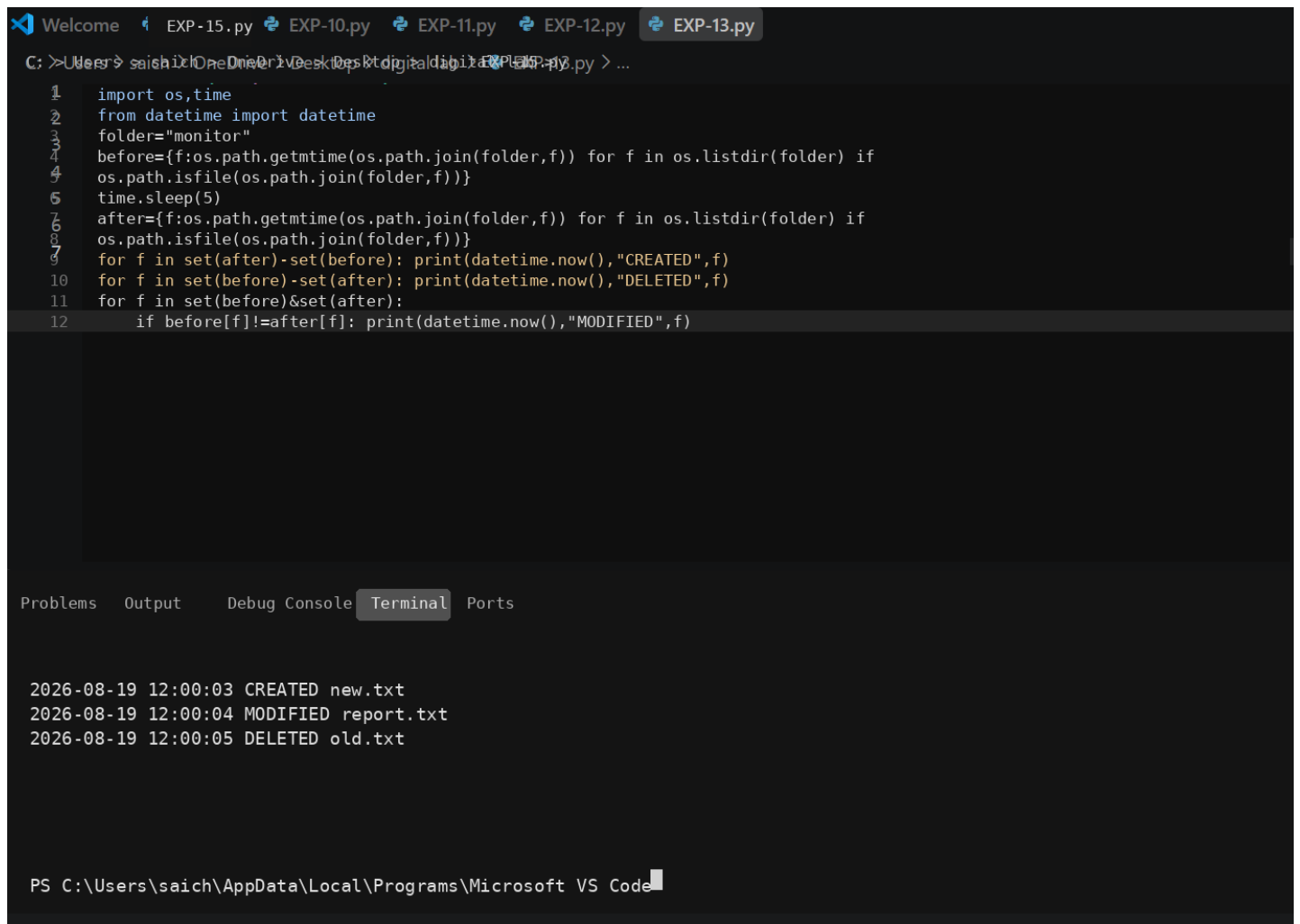
before = snapshot()
time.sleep(5)
after = snapshot()

for name in sorted(set(after) - set(before)):
    print(datetime.now(), "CREATED", name)

for name in sorted(set(before) - set(after)):
    print(datetime.now(), "DELETED", name)

for name in sorted(set(before) & set(after)):
    if before[name] != after[name]:
        print(datetime.now(), "MODIFIED", name)
```

OUTPUT



The screenshot shows a Visual Studio Code editor with a Python file named `EXP-13.py` open. The script monitors a folder named "monitor" for file system changes. It uses `os.path.getmtime` to get the modification time of files and compares it between two snapshots, separated by a 5-second sleep. It prints "CREATED", "DELETED", and "MODIFIED" messages with the file name and the current timestamp.

```
1 import os,time
2 from datetime import datetime
3 folder="monitor"
4 before={f:os.path.getmtime(os.path.join(folder,f)) for f in os.listdir(folder) if
5 os.path.isfile(os.path.join(folder,f))}
6 time.sleep(5)
7 after={f:os.path.getmtime(os.path.join(folder,f)) for f in os.listdir(folder) if
8 os.path.isfile(os.path.join(folder,f))}
9 for f in set(after)-set(before): print(datetime.now(),"CREATED",f)
10 for f in set(before)-set(after): print(datetime.now(),"DELETED",f)
11 for f in set(before)&set(after):
12     if before[f]!=after[f]: print(datetime.now(),"MODIFIED",f)
```

The terminal output shows the following log entries:

```
2026-08-19 12:00:03 CREATED new.txt
2026-08-19 12:00:04 MODIFIED report.txt
2026-08-19 12:00:05 DELETED old.txt
```

The status bar at the bottom indicates the file path: `PS C:\Users\saich\AppData\Local\Programs\Microsoft VS Code`.

RESULT

The program records observable file-system changes and produces an activity summary.